

SIGNIFICANCE

Role of this project within the African Colorectal Cancer Network and Disparities And Cancer Epidemiology (AfriCAN DANCE) Study: The Domestic Project of this application is the “translational engine” of AfriCAN DANCE. While the three foreign components operationalize case ascertainment and biospecimen collection across West and East Africa, the Domestic Project is where (1) the full complement of somatic and transcriptomic data will be integrated, (2) ancestry-enriched therapeutic vulnerabilities will be identified, and (3) candidate targets will be experimentally validated. The scientific rationale for ancestry-inclusive CRC genomics, the limitations of European-ancestry-focused reference databases, and evidence from other cancers establishing that African tumor genomes can uncover novel drivers are established in the Overall Research Strategy. This section focuses on the scientific opportunity that the DANCE cohort creates in the context of our multi-continental design and on why the translational pipeline from somatic discovery to organoid-based functional validation is essential to converting genomic findings into clinically actionable outcomes.

The somatic mutational landscape of CRC is clinically informative but incompletely characterized, particularly across ancestries: The somatic mutational landscape of CRC has been extensively characterized in predominantly European-ancestry cohorts and has yielded clinically actionable insights that now directly guide treatment decisions. Approximately 80–85% of CRCs are microsatellite stable (MSS) with chromosomal instability, 10–16% are microsatellite instable (MSI) due to mismatch repair deficiency (MMRd), and 1–2% are ultramutated due to somatic *POLE* mutations¹². MSI is associated with better response to immune checkpoint inhibitors, whereas *KRAS*, *NRAS*, and *BRAF* mutations predict poor response to EGFR-targeted therapies. Beyond these binary biomarkers, mutational signature analysis¹³ offers potential for stratifying CRC based on genetic and epigenetic alterations for precision oncology, identifying therapeutic vulnerabilities, and evaluating drug efficacy¹⁴. Mutations in *WNT*, *EGFR*, and *TGFβ* pathway genes, copy number variations, and the COSMIC SBS44 (MMR) signature have been linked to survival outcomes, underscoring that the clinical utility of mutational profiling in CRC extends well beyond the handful of biomarkers currently incorporated into standard-of-care¹².

Ancestry-specific somatic and transcriptomic differences in CRC have direct and underexplored therapeutic implications: For nearly every common cancer where race has been studied, significant differences in tumor subtype and somatic mutation profile explain at least part of the documented racial disparities in prognosis^{6–11}. The direct therapeutic consequences of these differences have now been demonstrated across multiple cancer types in ways that could not have been anticipated without ancestry-inclusive sequencing. Ancestry-stratified molecular profiling across multiple tumor types demonstrated that patients of African ancestry harbor distinct frequencies of actionable alterations, including enrichment of DNA repair pathway defects and immune pathway activation, that suggest divergent druggable target landscapes compared to patients of European ancestry, with direct implications for treatment selection¹⁵. In prostate cancer, the identification of elevated homologous recombination deficiency and DNA repair gene mutations in African-ancestry tumors opened a clinically testable hypothesis that PARP inhibitors may disproportionately benefit men of African descent, a hypothesis that could not have been formulated without ancestry-inclusive sequencing and that is now being evaluated in prospective trials¹⁶. These examples establish a generalizable principle: ancestry-inclusive genomics does not merely describe molecular differences, it generates testable therapeutic hypotheses that are directly actionable in the clinic.

Emerging evidence demonstrates that the somatic mutational spectrum of CRC differs meaningfully by race in ways directly relevant to treatment selection and outcomes, yet these differences are incompletely characterized and their therapeutic implications remain largely unaddressed. The most consistent and clinically consequential finding is that tumors from African-ancestry cases are approximately 10% more likely to contain *KRAS* mutations, which confer elevated mortality and resistance to EGFR inhibition^{17,18}, and a genomic profiling study of 324 cancer-related genes among over 46,000 CRC tumors confirmed significantly higher rates of *KRAS G12D*, *KRAS G13*, *APC*, and *PIK3CA* alterations in patients of African descent compared with European descent¹⁹. This study was limited by a lack of any demographic, clinico-pathologic or survival data. Conversely, AA patients have approximately 50% lower odds of harboring *BRAF V600E* mutations, which define a clinically aggressive CRC subtype for which *BRAF*-targeted combination regimens are now standard of care, meaning that treatment algorithms developed around European-ancestry mutation prevalence may be less applicable to AA patients, while the therapeutic implications of elevated *KRAS* burden in this population remain inadequately studied. Lower prevalence of MSI-H tumors in AA patients is also supported by recent evidence, a finding of particular consequence given that MSI status is now the primary biomarker guiding immune checkpoint inhibitor use in CRC²⁰. Critically, the evidence base for these racial differences derives almost entirely from targeted hotspot

panels rather than systematic whole-exome or whole-genome characterization, meaning that ancestry-related differences in rare drivers, mutational signatures, structural variants, and copy number alterations remain essentially uncharacterized. The Domestic Project addresses this gap directly by analyzing whole-exome sequencing and transcriptomic profiling across AA, Ghanaian, Ethiopian, and NHW CRC patients, providing the first comprehensive, ancestry-aware somatic catalog capable of moving beyond the handful of hotspot mutations currently studied toward a complete picture of how ancestral background shapes the CRC mutational landscape and its clinical implications.

Functional validation in ancestry-matched organoids is essential to distinguish drivers from passengers and predict drug response: The translational gap between genomic discovery and clinical application in CRC has been substantial. Patient-derived organoids (PDOs) have emerged as the gold standard platform for bridging this gap, faithfully recapitulating the genomic and phenotypic features of primary tumors and enabling direct drug sensitivity testing in a system that preserves inter-patient variability²¹. Drug sensitivity testing in CRC PDO biobanks consistently reveals that somatic genotype partially explains inter-patient variability in response to both standard-of-care and targeted agents^{22–24} but this genotype-to-drug-response relationship is precisely where ancestry matters most. If ancestry-specific somatic variants that stratify drug response are systematically underrepresented in existing biobanks, then genotype-to-drug-response relationships derived from those biobanks will have unknown and potentially misleading applicability to AA patients. Critically, existing CRC PDO biobanks have been built almost entirely from tumors of European ancestry, meaning the functional validation infrastructure carries the same ancestry bias as the discovery datasets it was designed to interrogate. A global analysis of patient-derived cancer models (all sites) shows, for patient-derived models overall: 48.3% no race/ethnicity reported; of those reported, White 37.5%, Asian 10%, AA 3.8%, Hispanic 0.4%²⁵. This suggests under-reporting and white overrepresentation in PDO resources used for oncology generally. The Domestic Project directly addresses this by establishing the first PDO biobank derived from AA CRC tumors, enabling functional validation experiments, including drug sensitivity testing across a panel of candidate inhibitors in an ancestry-matched experimental system. This is a prerequisite for generating preclinical evidence that is scientifically credible and clinically translatable for the patient population bearing the greatest CRC burden in the United States. The precedent from breast and prostate cancer, where ancestry-inclusive sequencing generated therapeutic hypotheses now under prospective clinical evaluation, has demonstrated that closing this loop in CRC is both scientifically achievable and clinically urgent.

INNOVATION

This project advances the current state of the science in ways that are distinct from and additive to the innovations described in the Overall Research Strategy, which covers design-level and infrastructure-level innovations shared across all components.

- 1. First integrated somatic and transcriptomic atlas of CRC across African-descendant populations.** Prior studies of racial differences in CRC molecular profiles have examined somatic mutations or gene expression in isolation, in single populations, or in datasets not designed for cross-ancestry comparison. The Domestic Project will generate the first study to systematically integrate WES-derived somatic mutation data with transcriptomic profiles across AA, Ghanaian, Ethiopian, and NHW CRC patients in a single, comparably ascertained, carefully annotated cohort. Somatic mutations identify candidate drivers, but transcriptomic data are required to determine which mutations produce consequential downstream pathway dysregulation and which represent the expression-level vulnerabilities most amenable to pharmacological intervention. No existing dataset or analytical effort has attempted this integration across African-descendant populations at this scale or with this degree of epidemiologic annotation.
- 2. A novel bioinformatic pipeline for ancestry-enriched druggable target nomination.** The Domestic Project will develop and apply a purpose-built computational pipeline that combines statistical association testing, machine learning-based feature selection, pathway enrichment analysis, and computational druggability assessment to nominate and rank therapeutic targets enriched in African-descendant CRC. This pipeline is innovative in three respects: it operates across multiple ancestry groups simultaneously rather than within a single population; it integrates somatic and transcriptomic signals rather than treating them as independent discovery streams; and it incorporates clinical outcome data, including survival and treatment response to prioritize candidates with demonstrated prognostic and therapeutic relevance rather than statistical association alone. The output will not be a list of differentially expressed genes, but a ranked, multi-criterion prioritization of candidates with the highest prior probability of functional validation and clinical actionability. This refined, data-driven computational approach allows for the repurposing of drugs already in

development to provide an efficient strategy for therapeutic discovery aimed at rapid translation.

3. **First ancestry-matched CRC organoid biobank and drug sensitivity testing framework.** The Domestic Project will establish the first PDO biobank derived from AA CRC tumors, filling a foundational gap in the CRC functional genomics infrastructure. This resource will be used to test the top candidate inhibitors nominated by the bioinformatic pipeline across paired AA and NHW PDOs, generating the first empirical, ancestry-matched genotype-to-drug-response data in CRC. The drug sensitivity testing framework is innovative in its comparative design, determining whether AA-derived PDOs are differentially sensitive to candidates relative to NHW-derived PDOs and directly testing the hypothesis that ancestry-enriched somatic and transcriptomic features translate into ancestry-enriched therapeutic vulnerabilities. This framework will serve as a replicable template for ancestry-aware functional validation in CRC and other cancer types. The organoid biobank will persist beyond the grant period as a shared scientific asset, enabling the research community to interrogate AA CRC biology, test emerging therapeutic hypotheses, and develop ancestry-calibrated predictive models in an experimental platform that currently does not exist.

APPROACH

Overall Design

The proposed research will address critical gaps in our knowledge of CRC by answering these key questions: What is the ancestry-specific somatic mutational landscape of CRC across AfriCAN DANCE populations? Which oncogenic drivers represent actionable targets enriched in African-descendant tumors? Do these drivers confer functional growth advantages and distinct therapeutic vulnerabilities in patient-derived organoid models? To answer these questions, this proposal leverages the infrastructure, biospecimens, and bioinformatics resources of the ongoing DANCE Colorectal Cancer Study (NIH/NCI

R01CA259420, MPI Rozek, Purrington). Using both banked formalin-fixed paraffin-embedded (FFPE) DANCE tumor specimens and prospectively collected samples from ongoing recruitment in Africa, we will conduct an integrated three-platform genomic study encompassing whole-exome sequencing, bulk RNA-seq, and patient-derived organoid functional validation across four ancestry groups: AA, NHW, Ghanaian, and Ethiopian. Our preliminary work using targeted DNA sequencing of 290 genes in DANCE samples (described below in Aim 1 Preliminary Data) demonstrates the feasibility of this approach and reveals biologically informative patterns of mutational profile similarities by genetic ancestry. Aim 1, led by Drs. Rozek, Purrington and Mabvakure, will define the ancestry-specific somatic mutational landscape using whole-exome sequencing. Aim 2a, led by Drs. Robbins and Wynn will identify druggable oncogenic drivers using bulk RNA-seq integrated with the expanded DANCE dataset augmented by public data, and Aim 2b, led by Dr. Tong, will functionally validate the highest-priority targets and assess therapeutic vulnerabilities using patient-derived and engineered organoid models. These aims form a fully integrated discovery-to-validation pipeline designed to produce an ancestry-aware, functionally validated map of druggable CRC targets across the spectrum of African-descendant populations.

Specific Aim 1: Define the landscape of somatic mutations in CRC among African descent populations using whole exome sequencing.

Preliminary Studies

The **Disparities and Cancer Epidemiology (DANCE)** cohort (R01CA259420, MPIs Rozek & Purrington) is a population-based cohort of 1,158 AA and NHW invasive CRC cases. All participants completed baseline questionnaires covering demographics, comorbid diagnoses, family history of cancer, lifestyle and health behaviors (smoking, physical activity, alcohol use), dietary factors, and medication use. Clinical data were obtained through the Metropolitan Detroit Cancer Surveillance System registry and the Louisiana Tumor Registry. ~71% provided blood/saliva and ~65% allowed access to tumor tissue. WES data (available & in

Table 1. Molecular and clinical characteristics of the cohorts that were sequenced.

Characteristic	GHA N = 21†	ETH N = 33†	AA N = 43†	NHW N = 53†	P value
MMR Phenotype					
dMMR	5 (31.3%)	1 (3.0%)	4 (25.0%)	4 (9.3%)	p = 0.0132
pMMR	11 (68.8%)	32 (97.0%)	12 (75.0%)	39 (90.7%)	
BRAF Status					
Mutant	2 (10%)	0 (0%)	0 (0%)	2 (4%)	p = 0.0895
Wild Type	19 (90%)	33 (100%)	42 (100%)	49 (96%)	
KRAS Status					
Mutant	1 (5%)	15 (45%)	18 (43%)	19 (37%)	p = 0.0053
Wild Type	20 (95%)	18 (55%)	24 (57%)	33 (63%)	

† n (%)

N = total samples available; n = samples tested; P value calculated using Fisher's Exact test.

GHA = Ghana; ETH = Ethiopia; AA = African American; NHW = Non-Hispanic White; MMR = mismatch repair; dMMR = deficient MMR; pMMR = proficient MMR.

progress) available for this project will include 300 AA and 100 NHW cases.

Utilizing targeted DNA sequencing data from 150 tumors (21 Ghanaian, 33 Ethiopian, 43 AA, and 53 NHW), we developed bioinformatics pipelines to characterize the mutational landscape of tumors using DNA sequencing data (manuscript submitted, under review). Overall, 9% NHW, 15% AA, 33% Ghanaian, and 2% Ethiopian cases were MMR-deficient (dMMR). The *BRAF* V600E mutation, commonly associated with poor prognosis and MSI-high tumors, was absent in AA and Ethiopian cases but was detected in 4% (2/53) of NHW tumors and 10% (2/21) of Ghanaian tumors (Table 1). For *KRAS* mutations,

43% of AA tumors carried mutations at codon 12/13, compared to 45% in Ethiopians, 37% in NHWs, and 5% in Ghanaians (Table 1). The somatic mutational landscape across study populations revealed significant differences, suggesting that the molecular drivers of oncogenesis vary considerably by population (Table 2). While *APC* mutations were nearly ubiquitous in the AA (93.0%) and NHW (92.5%) cohorts, their prevalence was significantly lower in the Ethiopian (60.6%) and Ghanaian (38.1%) groups ($p < 0.001$). Ghanaian and AA cohorts showed a higher frequency of *CTNNB1* mutations (25.6% and 19.0%, respectively) compared to the NHW (1.9%) and Ethiopian (9.1%) groups ($p = 0.0042$). Similarly, *RNF43* mutations showed significant variation ($p = 0.0047$), reaching their highest mutational frequency in the NHW cohort. *RNF43* was mutated in 73.6% of NHW samples, a rate nearly double that seen in the other populations. *PIK3CA* and *TP53* mutations showed significant variation ($p = 0.0002$ and $p = 0.0057$, respectively), with both genes reaching their highest mutational frequency in the AA and NHW cohorts, respectively. The AA cohort demonstrated a notably high *PIK3CA* mutational prevalence (39.5%) compared to the Ethiopian (3.0%) and Ghanaian (4.8%) cohorts, representing a highly significant distribution across the study groups ($p = 0.0002$). **Characteristics were grouped based on geography (Africa: Ghana & Ethiopia, US: AA & NHW) or by genetic relatedness (Afro-centric: AA & Ghanaian, Eurocentric: NHW & Ethiopian), suggesting there might be a mix of geography and ancestry factors at play.**

The majority of variants across all populations and tumor mutation burden (TMB) statuses were classified as “unknown” (grey bars), ranging from 87% in ETH TMB-low (TMBL) to 92.4% in AA TMB-high (TMBH) (Figure 1a). The most “unknowns” were identified among AA and GHA tumors compared to NHW and ETH samples across TMBL and TMBH groups. The proportion of pathogenic variants was notably higher among AA (4.5% and 1.5%) and NHW (5.1% and 1.6%) across TMBL and TMBH tumors, respectively. This pattern was also observed for variants with unknown clinical significance (VUS) among TMBH samples. In contrast, the proportion of pathogenic variants were 1.3% and 1.5% among Ethiopian and Ghanaian TMBL tumors respectively, and 0.8% among Ethiopian TMBH tumors. Given the high prevalence of VUS, we utilized PolyPhen to predict functional impact (Figure 1b). While the majority of VUS were predicted to be benign (40.2%-59.8%), a substantial fraction was predicted to be deleterious. Interestingly, TMBH tumors displayed a shift in functional distribution; compared to their TMBL counterparts, they exhibited a reduction in the proportion of benign variants and an increase in variants with “unknown”, “possibly damaging”, and “probably damaging” functional predictions. Higher proportions of “possibly/ probably damaging” variants were identified in AA and NHW TMBH tumors, as well as the lowest proportions of unknown variants. Seventy-four percent (74%) of the variants predicted to be probably damaging were absent in COSMIC, and their potential impact on the tumor biology of the different populations remains unexplored. **Taken together, these results suggest either distinct tumor biology in individuals of African descent compared with those of European descent or that variants in tumors from individuals of European descent are better characterized than those in other populations.**

Aim 1 methods

Study population: We will use biospecimens and questionnaire/clinical data from 1,400 AfriCAN DANCE

Table 2. Somatic mutation rates in key colorectal cancer genes.

Gene	AA	ETH	GHA	NHW	P value (Chi-sq)	Pathway impacted
<i>Wnt Pathway</i>						
<i>APC</i>	40 (93.0%)	20 (60.6%)	8 (38.1%)	49 (92.5%)	<0.0001*	<i>Wnt</i>
<i>CTNNB1</i>	11 (25.6%)	3 (9.1%)	4 (19.0%)	1 (1.9%)	0.0042*	<i>Wnt</i>
<i>FBXW7</i>	4 (9.3%)	3 (9.1%)	0 (0.0%)	4 (7.5%)	0.5599	<i>Wnt</i>
<i>RNF43</i>	19 (44.2%)	15 (45.5%)	8 (38.1%)	39 (73.6%)	0.0047*	<i>Wnt</i>
<i>MAPK & PI3K Pathways</i>						
<i>NRAS</i>	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (1.9%)	0.6057	<i>MAPK & PI3K</i>
<i>PIK3CA</i>	17 (39.5%)	1 (3.0%)	1 (4.8%)	11 (20.8%)	0.0002*	<i>MAPK & PI3K</i>
<i>TGF-Beta Pathway</i>						
<i>SMAD4</i>	1 (2.3%)	3 (9.1%)	2 (9.5%)	8 (15.1%)	0.2055	<i>TGF-Beta</i>
<i>TGFBR2</i>	4 (9.3%)	6 (18.2%)	2 (9.5%)	7 (13.2%)	0.671	<i>TGF-Beta</i>
<i>Other Key Genes</i>						
<i>TP53</i>	27 (62.8%)	15 (45.5%)	8 (38.1%)	40 (75.5%)	0.0057*	Other key genes

* Statistically significant ($p < 0.05$); P value calculated using Chi-square test.
AA = African American; ETH = Ethiopian; GHA = Ghana; NHW = Non-Hispanic White.

participants (AfriCAN: Ghana n=600, Ethiopia n=400; DANCE cases with paired tumor-normal WES available: US AA n=300, US NHW n=100). The US-based DANCE study is described above and in the Overall Research Strategy, and individual AfriCAN studies are described in Foreign Projects 1-3 Research Strategies. Samples obtained from studies include blood or saliva as a source of germline DNA and FFPE tumors.

Biospecimen processing, histopathology, and WES: All AfriCAN blood samples, saliva kits, and tumors will be processed centrally in Dr. Rozek's laboratory at GU for consistency with DANCE methodology, including histopathologic review, macrodissection of unstained tissue sections with >70% tumor cellularity, and DNA/RNA extraction. CRC tumors will undergo histopathologic review for well-established clinically relevant features such as differentiation, tumor necrosis, tumor infiltrating lymphocyte counts, mucinous composition, Crohn's-like reaction, and tumor budding. DNA will be extracted using the QIAamp DNA FFPE Tissue kit (Qiagen, Germany) with the QIASymphony automated extraction system. WES will be performed using the same methodology as for the DANCE cohort. DNA will be assessed using ThermoFisher Qubit 1x dsDNA High Sensitivity Assay and Agilent TapeStation genomic DNA reagents. Libraries will be prepared using Twist Library Preparation EF kit 2.0 per manufacturer's protocol with 50ng DNA input. Libraries were checked for quantity by Qubit hsDNA and LabChip, and pooled for capture as 16 samples per pool. Each pool will be hybridized to the Twist Exome 2.0, which covers 36.5 Mb of human protein coding regions, using manufacturer's protocols. The purified captured DNA will then be PCR amplified. Final capture pools will be checked for quality and quantity by Qubit hsDNA and LabChip and sequenced on the Illumina NovaX Paired-end 150bp, according to manufacturer's recommended protocols. Target coverage is 300X for tumor DNA and 100x for germline DNA.

Somatic variant calling and annotation: Fastq files of tumor-normal matched WES data for CRC tumors will be mapped to both the human reference genome (GRCh38) and the human pangenome reference²⁶, which integrates 47 phased diploid assemblies representing diverse global ancestries and captures 119 million base pairs of sequence absent from GRCh38, directly addressing the reference bias introduced when calling variants in non-European populations using a predominantly European-ancestry reference. Somatic variant calls from both alignments will be compared to quantify reference-introduced bias, and pangenome-specific variants identified in African-ancestry samples will be prioritized as candidate ancestry-enriched drivers. Quality metrics (per-base sequence quality, GC content, sequence duplication levels, overrepresented sequences) will be assessed using FastQC (v0.11.9), and summary statistics (total read counts, mismatch rates, error rates, mapped/unmapped read proportions) will be assessed using SAMtools. Quality results will be aggregated using

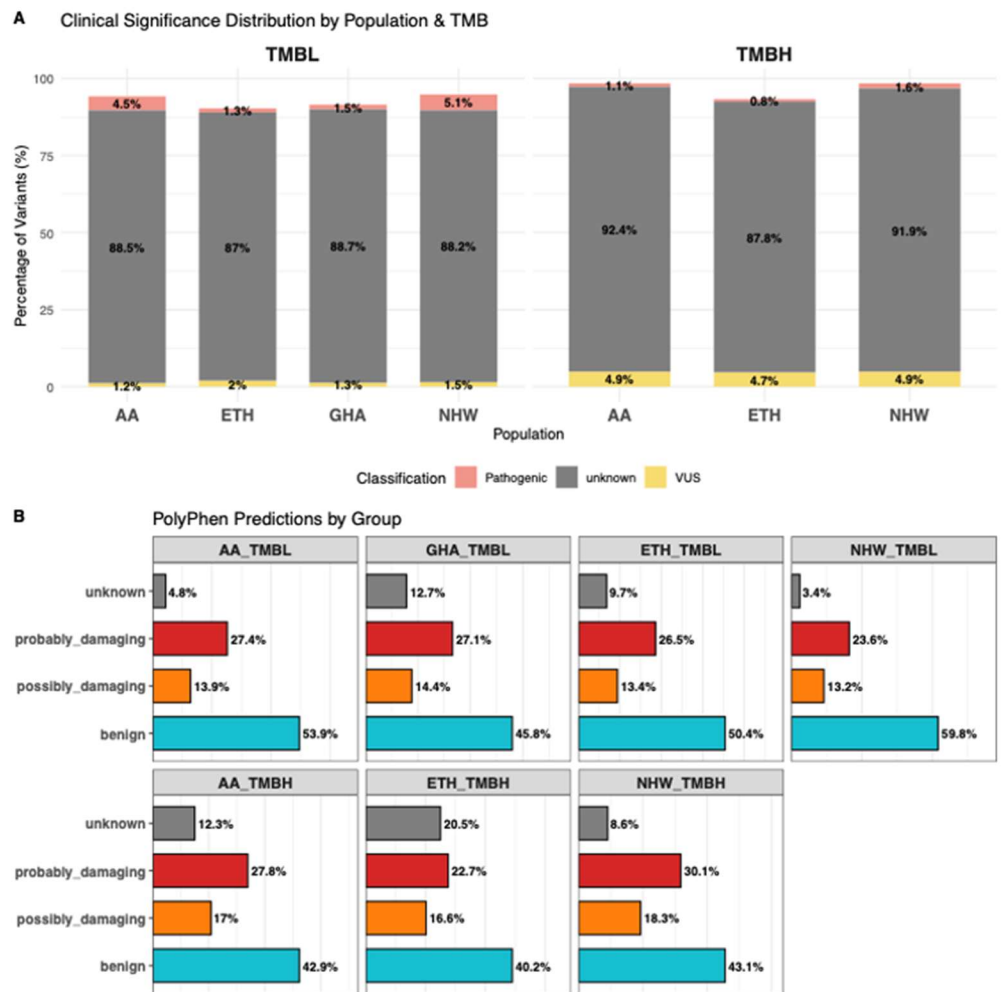


Figure 1. Clinical classification and computational functional assessment of somatic variants. (A) Stacked bar charts displaying the proportion of somatic variants classified by clinical significance (ClinVar) across African American (AA), Ghanaian (GHA), Ethiopian (ETH), and Non-Hispanic White (NHW) populations, stratified by TMB-High and TMB-Low status. (B) Functional impact predictions (PolyPhen) specifically for the subset of variants with "Unknown" clinical significance from Panel A, categorized as Benign, Possibly Damaging, or Probably Damaging.

MultiQC (v1.28), which consolidates metrics into a single interactive report. Somatic variant calling on tumor-normal matched pairs will be performed using GATK Mutect2 within the nf-core/sarek pipeline. Tumor purity and ploidy will be estimated using PURPLE to account for their influence on variant allele frequencies and tumor mutation burden (TMB) estimates, and variant calls will be adjusted accordingly. Resulting VCFs will be annotated using the Broad Institute's GATK Funcotator tool against gnomAD, ExAC, the 1000 Genomes Project, ClinVar, and COSMIC databases, generating mutation annotation files (MAFs). Functional impact scores will be estimated using SIFT and PolyPhen. Microsatellite instability (MSI) status will be estimated from WES data using MSISensor2, and MSI tumors will be analyzed separately given their characteristically elevated TMB. Tumors will be classified as MIS or microsatellite stable (MSS) based on the MSISensor2 score, where a score of 3.5 or above is classified as MSI-H and below 3.5 as MSS, consistent with established thresholds in the literature.

Genetic ancestry estimation: VCFs for germline variant discovery will be generated from the nf-core/sarek pipeline using GATK HaplotypeCaller. PLINK2 will be used to filter for high-quality biallelic single-nucleotide polymorphisms and perform linkage disequilibrium pruning ($r^2 < 0.1$). Principal component analysis will be performed using PLINK2 to confirm population clustering and identify outliers prior to supervised ancestry estimation. The dataset will be merged with the 1000 Genomes Project Phase 3 reference panel. Population structure will be characterized using a maximum likelihood-based approach implemented in ADMIXTURE (v1.3.0). A supervised clustering model with K=5 ancestral populations (African, European, East Asian, South Asian, and Admixed American) will be used to estimate individual admixture proportions.

Mutational signature analysis: Mutational signature analysis will be performed using SigProfilerExtractor to identify the COSMIC mutational signatures¹³ operative in CRC tumors across the Ghanaian and Ethiopian populations. This analysis will characterize the contribution of known signatures associated with environmental and lifestyle exposures, including those linked to diet, aflatoxin exposure, and defective DNA mismatch repair, enabling direct connections to be drawn between the genomic data and the exposure patterns relevant to each population. Signature activities will be compared across geographic and ancestry groups to identify population-specific patterns of mutagenic exposure.

Tumor mutation burden & subtyping: TMB and top mutated genes across all comparison categories will be estimated using MAFtools in R. TMB will be calculated by dividing the number of somatic non-synonymous mutations by the estimated size of the coding region of the genome. Tumors will be classified as TMB-high (TMB-H, ≥ 10 mutations per megabase) or TMB-low (TMB-L, < 10 mutations per megabase) based on established clinical guidelines. TMB classification will be integrated with MSI status to subtype tumors into four biologically distinct categories: TMB-H/MSI-H, TMB-H/MSS, TMB-L/MSI-H, and TMB-L/MSS. This combined classification will allow characterization of the distribution of biologically and clinically distinct tumor subtypes across Ghanaian and Ethiopian populations, and will inform analyses of how genetic ancestry and geographic factors influence tumor immunogenicity and potential responsiveness to immunotherapy.

Analytic approach & driver gene identification: Mutational data will be integrated with race, geography, age, and sex to estimate associations between mutational signatures and these factors. Analyses will be performed in R using the Bioconductor MAFtools package²⁷. Minor allele frequencies (MAFs) will be imported into R and variants will be filtered based on a minimum sequencing depth of 20x and a $MAF < 1\%$ in each population database (ExAC, gnomAD, 1000 Genomes Project).

Recent large-scale whole-genome sequencing studies have identified over 250 putative driver genes, many not previously implicated in CRC, but these analyses remain predominantly European in ancestry. In the AfriCAN DANCE cohort, we will systematically evaluate both well-established, frequently mutated CRC driver “mountains” (*APC*, *BRAF*, *CTNNB1*, *FBXW7*, *KRAS*, *MLH1*, *MSH2*, *MSH6*, *NRAS*, *PIK3CA*, *PMS2*, *RNF43*, *SMAD4*, *TGFBR2*, *TP53*) and the broader landscape of putative “hill” genes identified in contemporary genomic studies^{12,28,29}, testing the hypothesis that the relative frequency and functional impact of hill genes differs by African ancestry. Critically, genes that appear as hills in TCGA-derived analyses may represent mountains in African-ancestry tumors, with implications for targeted therapy and risk stratification.

WES data from AfriCAN DANCE participants (n=1,400 tumor-normal pairs: 300 US AA, 100 US Non-Hispanic White (NHW), 600 Ghanaian, 400 Ethiopian) will be processed through a driver gene identification pipeline modeled on the approach used in the UK 100,000 Genomes Project CRC analysis¹¹. Mutations will be analyzed using seven complementary driver identification methods implemented through the IntOGen pipeline: dNdScv, cBaSE, OncodriveCLUSTL, smRegions, HotMAPS, OncodriveFML, and MutPanning. These methods collectively test for orthogonal signals of positive selection, including excess non-synonymous mutation burden

(corrected for trinucleotide context), spatial clustering of mutations along the protein sequence and in 3D structure, and enrichment of functionally impactful variants as scored by Combined Annotation Dependent Depletion (CADD). To account for the confounding effects of hypermutation on background rate estimation, all analyses will be stratified by microsatellite stability status (MSS, MSI, and *POLE*-mutant), with hypermutated samples handled via iterative subsampling as described by Nunes et al.¹² A gene will be considered a putative driver if it achieves an FDR q-value < 0.05 with support from at least two independent methods; genes in the COSMIC Cancer Gene Census will be evaluated at relaxed thresholds (CGC q-value < 0.25) to preserve sensitivity for known drivers present at lower frequencies in this population. Identified driver genes will be classified by mechanism of action (activating, loss-of-function, or ambiguous) using dNdScv mutation distribution patterns and cross-referenced against the established CRC driver landscape^{12,28,29} to distinguish confirmed mountains, candidate hills, and novel African ancestry-associated drivers.

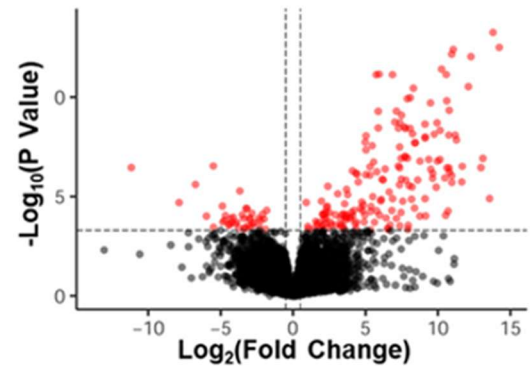


Figure 2. The transcriptome of CRC in patients with African ancestry is distinct. Volcano plot showing the differential gene expression associated with African ancestry. Red symbols indicate genes with statistically significant differential expression as determined by a likelihood ratio test ($p < 0.05$) and a $\log_2$ (fold change) of more than the absolute value of 0.5. Black symbols indicate genes with no significance and low fold change.

Statistical power for driver gene discovery: Power for somatic driver gene identification was assessed using a binomial detection framework, defining a gene as detectable when the probability of observing mutations in at least two independent tumor samples exceeds 80%. At $N=1,400$ tumor-normal pairs, we achieve 80% power to detect driver genes present at a true frequency as low as **0.2%**, with expected mutation counts exceeding 6 at 0.5% frequency and exceeding 13 at 1.0% frequency. For well-established CRC mountains [*APC* (~80%), *TP53* (~60%), *KRAS* (~45%)] detection power is effectively 100% in all subgroups including Ethiopia alone ($N=400$). For lower-frequency hill genes such as *PIK3CA* (~18%), *BRAF* (~10%), and *SMAD4* (~10%), expected mutation counts in the full cohort are 233, 130, and 130, respectively, ensuring robust detection and stratified analysis by MSI status. Within-country subgroup analyses (Ghana $N=600$, Ethiopia $N=400$) remain powered to detect genes at frequencies as low as 0.5% and 0.9%, respectively.

Statistical analyses & power for comparing variants across categories: As for discrete categories such as population/ geography, we will apply a generalized linear model (GLM) with a binomial family to identify variants with significantly different frequencies between the two comparison groups. Spearman correlation will be used for continuous genetic ancestry analyses. Multiple testing correction will be applied using the Benjamini-Hochberg false discovery rate method, with a significance threshold of $FDR < 0.05$. Analyses will be structured across the following comparison categories:

1. **Within country/ geography:** Mutations in CRC tumors will be characterized by geographic origin, including Ethiopia and Ghana separately.
2. **Between countries/ geography:** Mutations will be compared across geographic groups to identify population-specific patterns, including Ghana vs. Ethiopia, Ghana vs. the United States, and Ethiopia vs. the United States.
3. **Genetic ancestry:** Ancestry estimations from ADMIXTURE will be applied in two ways:
 - a. **Continuous AFR ancestry proportions:** The proportion of African (AFR) ancestry in each tumor sample will be estimated using the 1000 Genomes Project reference panel, producing a continuous distribution ranging from 0 to 1 (0 to 100%).
 - b. **Categorical groups:** Tumors will be stratified into three groups based on AFR ancestry proportion using a quantile-based approach: Low AFR ($\leq 30\%$), Medium AFR (30% to 80%), and High AFR ($\geq 80\%$).

Power for pairwise comparisons of mutation frequencies was calculated using a two-sided two-proportion z-test ($\alpha=0.05$, Benjamini-Hochberg FDR-corrected), with baseline frequencies derived from TCGA CRC estimates for six representative driver genes spanning the full frequency range: *APC* (80%), *TP53* (60%), *KRAS* (45%), *PIK3CA* (18%), *BRAF* (10%), and *SMAD4* (10%). For the three primary cross-population comparisons, we have $\geq 80\%$ power to detect the following minimum absolute differences: Ghana ($N=600$) vs. Ethiopia ($N=400$): 6.5–9.2%; Ghana vs. US AA ($N=300$): 7.5–10.7%; Ethiopia vs. US AA: 8.4–11.8%, depending on baseline frequency. For a 10 percentage-point difference, power exceeds 85% for all three primary comparisons, exceeding 93% for *BRAF* and *SMAD4* in the Ethiopia vs. US AA comparison. For a 15 percentage-point difference, power exceeds 94% for all primary comparisons. The US NHW group ($N=100$) is included as an ancestry reference; pairwise comparisons involving this group are treated as exploratory (power 31–61% for a 10pp difference). For

continuous African ancestry proportion analyses using Spearman correlation (N=1,400), the study achieves 80% power to detect correlations as small as $r=0.08$ and 95% power at $r=0.10$, providing sensitivity for modest ancestry-associated effects on somatic mutation patterns.

Potential problems and alternative solutions: FFPE specimens from African sites may have variable DNA quality due to differences in fixation protocols and tissue handling. We have addressed this by implementing pre-shipment H&E quality review at each site, requiring >70% tumor cellularity, and using the QIASymphony automated extraction system validated for degraded FFPE inputs. For samples with insufficient DNA yield for standard WES, we will apply low-input library preparation at reduced input, which has been validated in the Rozek laboratory. We address divergence of African-specific germline variation from GRCh38 by mapping all samples in parallel to both GRCh38 and the human pangenome reference, comparing variant calls across alignments to quantify and flag reference-introduced artifacts. For variants flagged as potential false positives due to reference bias, we will apply gnomAD African population allele frequency filters and cross-reference against the COSMIC misclassification database to systematically distinguish true somatic mutations from African-specific germline variants incorrectly called as somatic. All driver analyses will be stratified by MSI status with hypermutated samples handled via iterative subsampling, preventing inflation of apparent driver signals in these subgroups. If MSI-H prevalence in AfriCAN cohorts differs substantially from TCGA priors, we will recalibrate background mutation rate models using country-stratified trinucleotide context correction.

Specific Aim 2. Identify therapeutic vulnerabilities in CRC patients with African ancestry.

Overview: Our preliminary analysis suggests that distinct patterns of gene expression exist within CRC patients with African ancestry, a subset of which are driven by somatic mutations enriched in this CRC subpopulation. We will use RNA sequencing (RNAseq) to determine the transcriptomes of 400 Ghanaian and Ethiopian CRC tumor samples combined with existing RNAseq tumor data (100 AA, 100 NHW). We will apply a bioinformatic pipeline that employs a series of statistical and machine learning approaches to identify druggable therapeutic vulnerabilities. Candidate drug targets will be prioritized based on their association with clinical outcomes, enrichment of expression, machine learning metrics, and participation in African ancestry-enriched CRC pathways. Functional assessment of candidates will then be performed using patient-derived organoid (PDOs), which serves as a high-fidelity model system for studying the underlying mechanisms of CRC. We will then compare the response of 30 PDOs derived from AA patients with 20 PDOs derived from NHW (all with % African ancestry quantified) using a range of concentrations of each inhibitor. Collectively, these studies will identify small molecules capable of the selective attenuation of CRC tumor growth in patients with African ancestry.

Preliminary data

To address gaps in our knowledge of CRC¹¹, we analyzed bulk RNAseq data using samples collected in the US, Ghana, and Ethiopia (Total n=77: AA=18, NHW=32, ETH=24, GHA=3). **This preliminary analysis was underpowered to detect differences between each of the four populations (which is the focus of this PF5 research project). These preliminary data demonstrate (1) feasibility of generating RNAseq data from samples from each country, (2) ability to detect ancestry-related differences in gene expression in these samples, and (3) successful implementation of our novel pipeline.**

Exploration of ancestry-related differences in the CRC transcriptome: Using the targeted DNA sequencing results discussed above, the proportion of African ancestry within each sample was quantified using ADMIXTURE³⁰, ranging from 5% to 85% (median=37%, IQR=26.5%). Our initial differential gene expression analysis models³¹ identified 156 differentially expressed mRNA transcripts between African ancestry (US AA, Ghana, Ethiopia) and NHW patients (**Fig. 2**). Gene Set Enrichment Analysis (GSEA) was used to quantify associated pathway activity, showing that E2F and MYC signaling as well as metabolic activity are correlated with African ancestry (**Fig. 3**). Conversely, our results suggest that epithelial mesenchymal transition is less prevalent in CRC patients with high degrees of African ancestry. Interestingly, MYC signaling is activated by the WNT and PI3K signaling cascades, which may be a function of the divergent frequency of mutations in these pathways discussed above (**Table 2**). To corroborate these results, we performed an analysis of the expression of regulons associated with specific transcription factors, using the DoRoTHEa database³² of transcription factor target genes and virtual inference of protein-activity by enriched regulon analysis³³. This analysis revealed enhanced activity of the transcription factors CDX2 and MYB, both of which have been previously linked with MYC signaling activity³⁴⁻³⁶. These findings further imply that transcriptional programs in CRC derived from patients of African ancestry are distinct as a result of signaling and metabolic activity.

Prediction of therapeutic vulnerabilities: To demonstrate our ability to identify oncogenic drivers in CRC patients

of African ancestry, we next employed our novel computational analysis. Elastic net regression³⁷ and random forest modeling³⁸ were used in an approach employing cross-validation³⁹. To ensure the rigor and reproducibility of our analysis, only features stable over >10 iterations were considered valid. The results of these two models were cross-referenced to produce a list of candidate oncogenic drivers. Consistent with Aim 1 preliminary data, gene products involved in immune signaling, particularly that cytokine signaling and the complement system, were enriched in CRC patients of African ancestry (Fig. 4). Interestingly, gene products involved in the activity of RHO family GTPases, which are frequently activated by PI3K signaling^{40–43}, were also identified by both machine learning algorithms. Importantly, an analysis of CRC patient samples in the TCGA found that 41 of these candidate drivers are associated with survival in AA but not NHW patients. Furthermore, 12 of these drivers are druggable and represent actionable opportunities for therapeutic intervention.

Generation of patient-derived CRC organoids: In a coordinated effort at HMH JTCC, we derived 120+ matched CRC patient tumor and normal colon organoids. We assessed the sensitivity of PDOs to 5-FU based frontline chemotherapies for CRC patients, which is still the current standard of care for CRC patients at HMH-JTCC. While normal, non-tumor organoids show no significant decrease in viability, CRC tumor organoids from the same patient show significant sensitivity to 5-FU based treatment with a 50% decrease in cell viability following 7 days of treatment, supporting our previous publication that CRC PDOs can reflect drug sensitivity and potentially model patient response to targeted therapy⁴⁴(Fig. 5). Consequently, we are currently following 5 patients undergoing 5-FU based treatment and reveal that PDO sensitivity to 5-FU treatments reflect CRC patient response. **We have extended this workflow to GU (P30CA051008) to capture a diverse cohort of CRC patients where from 2021 - 2024, 44.8% of the CRC cases were self-described as AA.** This collection will take advantage of an IRB-approved protocol at GU for cohort ascertainment and specimen collection.

Aim 2a methods: Identify therapeutic vulnerabilities via a bioinformatic pipeline that employs a series of statistical and machine learning approaches

RNA Sequencing: DNA/RNA co-extractions will be performed as described in Aim 1. RNA from 400 AfriCAN DANCE cases (200 Ghanaian and 200 Ethiopian, selected to represent the full range of clinical and pathologic characteristics in the cohort) will be processed and sequenced at the University of Maryland Sequencing Core. A minimum of 100 AA and 100 NHW DANCE participants with existing RNA-seq data will be included, selected

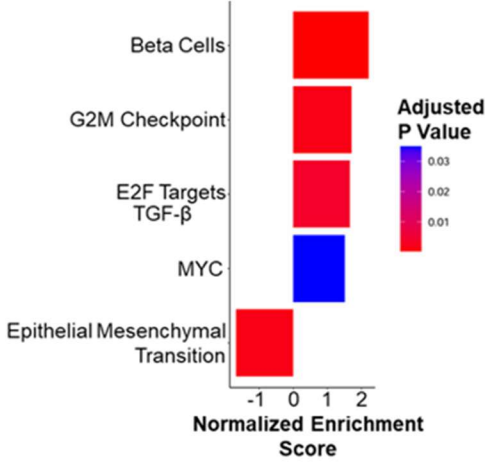


Figure 3. MYC and E2F signaling are correlated with African ancestry in CRC. A) Bar graph showing the top differentially regulated hallmark pathways, from the Molecular Signatures Database, associated with ancestry. Pathways are shown on the y-axis. The x-axis shows the normalized enrichment score and color indicates adjusted p value.

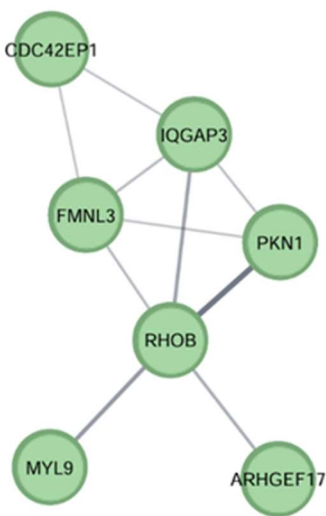


Figure 4. Machine learning algorithms predict RHO GTPase signaling effectors function as oncogenic drivers linked with African ancestry. Gene products selected by our bioinformatic pipeline were used to generate and cluster a network of functional interactions via the STRING database in order to identify enriched pathways.

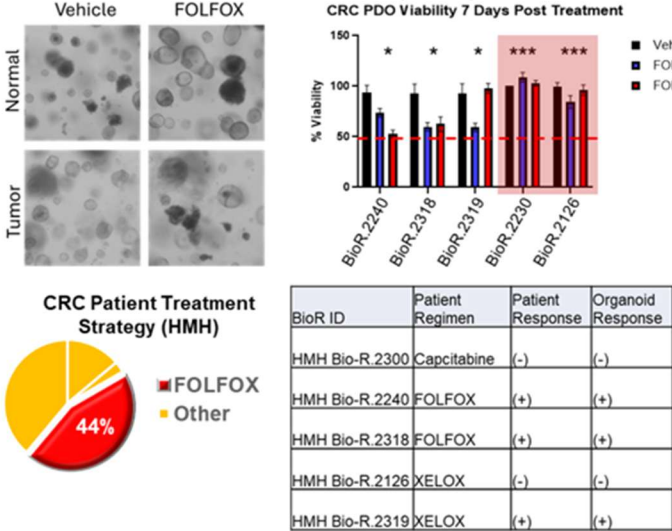


Figure 5. Ex vivo assessment of 5-FU based treatment in CRC PDOs. CRC patient matched normal and tumor organoids were treated with either vehicle or 5-FU based treatment (FOLFOX/FOLFIRI) to assess tumor sensitivity. FOLFOX is still the primary treatment strategy for CRC patients at HMH. Individual PDO lines show heterogenous responses to 5-FU treatment (* = sensitive; *** = resistant). Preliminary findings reveal that organoid responses correlated with patient responses to 5-FU based treatment

to frequency match African cases on age of onset category (<50, 50+), stage, and sex. Raw fastq files will be processed using STAR alignment/quantification software⁴⁵ to produce a matrix containing quantified count values for all detected genes. As in Aim 1, alignment quality will be assessed using FastQC⁴⁶. The presence of outliers and batch effects will be determined using several methods after count data are normalized using a variance stabilizing transformation, including principal component analysis, sample distance, and hierarchical clustering. To capture the full spectrum of African ancestry represented in our cohort rather than treating ancestry as a binary category, DNA sequencing data from Aim 1 will be used to estimate the proportion of African ancestry in each participant using ADMIXTURE³⁰ and RFMix⁴⁷, producing continuous ancestry proportion scores that will serve as the primary covariate in all transcriptomic analyses. These ancestry-quantified, batch-corrected RNA-seq data will then enter the ancestry-aware drug discovery pipeline described below (Fig. 6).

Identification of the CRC transcriptomic features associated with African ancestry: We will evaluate differential expression using the same framework as described for Aim 1 (within/between country, by ancestry). Likelihood ratio tests will be performed before Benjamini-Hochberg multiple testing correction⁴⁸ (statistically significant = $p < 0.05$, $\log(\text{fold change}) > 0.5$). Log (fold change) values will be shrunk using Bayesian estimation⁴⁹. We will use African ancestry estimates as continuous variables in GLMs using median-of-ratio transformed counts via DESeq2³¹. Models will be adjusted for potential confounders (stage, site, MSI, collection site, batch). We will then evaluate the enrichment of pathway activity in CRC patients within subgroup using GSEA⁵⁰ and the Molecular Signatures Database hallmark⁵¹, KEGG, REACTOME⁵², and Gene Ontology consortium gene set collections. These results will be corroborated by assessing transcription factor activity using virtual inference of protein-activity by enriched regulon analysis³³ and the DoRothEA database³². Associations between transcription factor activity and subgroup will then be assessed using linear regression controlling for appropriate confounders.

Prioritization of druggable oncogenic drivers in CRC patients of African ancestry: To identify candidate therapeutic vulnerabilities, we will employ two machine learning algorithms that predict the highest probability of functioning as oncogenic drivers specific to this demographic group. This includes some of the most commonly used approaches for cancer biomarker discovery^{53–56}, elastic net regression³⁷, as well as random forest modeling³⁸. These methods function through distinct processes with complementary sets of advantages and limitations, providing an orthogonal approach to candidate prioritization. Furthermore, cross-validation³⁹ will be used to ensure robustness. Model performance will be evaluated using the Jaccard similarity index⁵⁷ and the receiver operating characteristic⁵⁸, gold standards for reproducibility and accuracy, respectively. Only candidates that are stable over > 10 iterations of each method will be prioritized to guarantee the rigor and reproducibility of our analysis. Genes significantly associated with a particular subgroup, predicted to be oncogenic drivers by machine learning algorithms, and upregulated in statistical models will be assessed for druggability using the Drug Gene Interaction Database⁵⁹. MatrixEQTL⁶⁰ analyses will allow us to identify candidate drug targets with expression driven by mutations enriched in a particular subgroup. Cox proportional hazard models⁶¹ will be used to determine the association of the expression of this druggable subset of candidate molecular drivers with patient outcomes using data from the AfriCAN DANCE cohort. Publicly available datasets, including the TCGA⁶², will be used as external validation of these findings, taking into account the lack of sequencing depth and variant misclassification commonly seen in African-ancestry samples in these datasets.

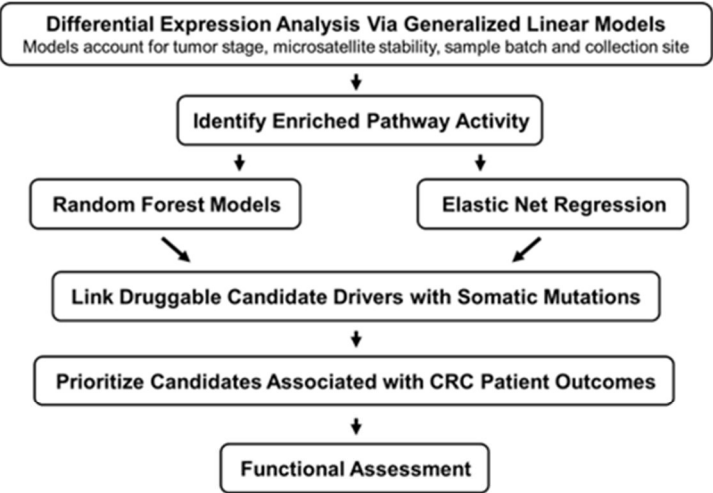


Figure 6. A schematic representation of our oncogenic driver identification and prioritization pipeline.

Table 3. RNA-seq Power Analysis: Ancestry-Stratified Druggable Target Discovery

Fold change	Power	Sample size required	
		Per group	Total (4 groups)
1.25-fold	70%	138	552
1.25-fold	80%	162	648
1.25-fold	90%	202	808
1.5-fold	70%	41	164
1.5-fold	80%	48	192
1.5-fold	90%	59	236
2-fold	70%	14	56
2-fold	80%	16	64
2-fold	90%	19	76

Parameters: BCV = 0.4 | FDR = 0.05 | m = 20,000 total genes | m1 = 2,000 estimated DE genes | lambda0 = 5 | equal group sizes (k = 1). BCV = 0.4 based on published estimates for heterogeneous CRC tumor bulk RNA-seq (Hart SN et al. *Bioinformatics* 2013; TCGA CRC cohort).

Power calculations: RnaSeqSampleSize⁷ provides genome-wide FDR-corrected sample size estimates accounting for multiple testing across the full transcriptome (**Table 3**). We assumed a biological coefficient of variation of 0.4 (consistent with published estimates for heterogeneous CRC tumor bulk RNA-seq data), FDR of 0.05, 20,000 genes tested, and a conservative estimate of 2,000 truly differentially expressed genes. Given the potential subtle regulatory influences rather than large binary shifts, we powered the study to detect a minimum 1.5-fold change. At 80% power and a 1.5-fold minimum detectable change, 48 samples per group (192 total samples) across the four ancestry groups are required.

Aim 2b methods: Functionally assess drug candidates using patient-derived organoid (PDOs)

Target African-specific CRC drivers ex vivo using PDO cultures: Thirty AA-derived CRC PDO lines will be established from patients at HMH-JTCC and LCCC. De-identified CRC patient resections will be viably preserved. 10% will be fixed and paraffin-embedded for histological analysis and validation of oncogenic molecular drivers and ancestry-enriched markers identified in Aim 2a. The remainder will be enzymatically digested and embedded in Matrigel for organoid derivation. To account for an expected 20–30% derivation attrition rate, we will attempt derivation from 40 AA patient resections to yield 30 viable, expandable lines for the screen. Patients with hereditary CRC predisposition syndromes (Lynch Syndrome, FAP) will be excluded. Twenty NHW-derived PDO lines will be drawn from the existing HMH-JTCC PDO bank. All 50 PDO lines will undergo WES prior to drug sensitivity testing to estimate African ancestry proportion using ADMIXTURE as described in Aim 1, characterize the somatic mutational landscape of each line, and determine variant carrier status for each of the top candidate ancestry-enriched drivers from Aim 1.

Variant-stratified primary drug sensitivity screen: All 50 PDO lines will be treated with the top 20 candidate inhibitors nominated by Aim 2a bioinformatic pipeline, plus vehicle control and current standard-of-care regimens (FOLFOX), across a range of concentrations spanning at least four orders of magnitude in 384-well plates. Treatment will continue for 14 days or until significant differences in organoid viability, growth rate, or morphology are observed, assessed by microscopy and CellTiter-Glo 3D Cell Viability Assay. Dose-response curves for each PDO-inhibitor combination will be generated using three-parameter non-linear regression, and potency (IC₅₀) and efficacy (Emax) calculated for each line. All lines will be passaged to a standardized passage number before screening, Matrigel lots will be pre-tested and held consistent across the full screen, and a validated CRC reference line will be included across all screening batches as an internal control. The primary analysis for each candidate inhibitor will compare drug sensitivity between PDO lines carrying the matched ancestry-enriched driver variant and lines that do not, using a mixed effects model with variant carrier status as the primary predictor and African ancestry proportion as a continuous covariate to test whether the variant predicts sensitivity. We will also test whether African ancestry proportion predicts sensitivity independently of variant carrier status.

CRISPR knock-in for low-frequency variant augmentation and isogenic validation: For low-frequency variants (n<4), CRISPR-based knock-in will be used to (1) augment the carrier group to at least 8 lines to increase statistical power and (2) provide isogenic mechanistic validation by introducing the variant into a controlled NHW genetic background. NHW-derived PDO lines confirmed by WES to be wild-type at the locus of interest will be selected for editing. Organoids

Table 4. Statistical power for variant-stratified PDO drug sensitivity screen

Variant	Freq AA	Freq NHW	Expected natural carriers (30 AA + 20 NHW)	Power (natural only)	Post-CRISPR augmentation strategy	Power (post-CRISPR)	CRISPR requirement
<i>High-frequency variants — natural carriers adequate, CRISPR not required</i>							
KRAS G12D/G13	43%	37%	21 carriers	93%	Not needed	93%	No CRISPR needed
PIK3CA	40%	21%	16 carriers	88%	Not needed	88%	No CRISPR needed
APC	38%	93%	30 carriers	94%	Not needed	94%	No CRISPR needed
RNF43	44%	74%	28 carriers	92%	Not needed	92%	No CRISPR needed
<i>Moderate-frequency variants — adequate power with 30 AA lines, CRISPR not required</i>							
CTNNB1	26%	2%	8 carriers	80%	Not needed	80%	No CRISPR needed
Novel drivers (est.)	25%	5%	9 carriers	83%	Not needed	83%	No CRISPR needed
<i>Low-frequency variants — CRISPR knock-in required in both 40-line and 50-line designs</i>							
Rare drivers (est.)	10%	2%	4 carriers	47%	8 carriers in NHW	80%	CRISPR required
SMAD4	2%	15%	4 carriers	47%	8 carriers in NHW	80%	CRISPR required
BRAF V600E	0%	4%	1 carrier	12%	8 carriers in NHW	80%	CRISPR required

Freq AA = variant frequency in African American CRC patients; Freq NHW = variant frequency in non-Hispanic White CRC patients.

Power estimated using two-sample t-test on IC₅₀ values assuming two-fold IC₅₀ difference, SD = 0.8 log units, α = 0.05, two-sided.

Post-CRISPR power assumes augmentation to 8 total carrier lines via knock-in of the specific variant into wild-type NHW-derived PDO lines.

Frequency estimates for KRAS, PIK3CA, APC, CTNNB1, RNF43, BRAF, and SMAD4 derived from Hembree et al. 2022 (Cancer Discovery) and African DANCE preliminary data. Novel African driver estimates are hypothetical based on analogous ancestry-enriched analyses.

will be dissociated to single cells and nucleofected with Cas9-guide RNA ribonucleoprotein complexes together with a single-stranded oligodeoxynucleotide HDR template encoding the specific variant of interest. Edited organoids will be re-embedded in Matrigel, expanded, and screened by Sanger sequencing and targeted deep sequencing to confirm successful knock-in. Off-target editing will be assessed at the top five predicted off-target sites by amplicon sequencing. Lines with off-target edits will be excluded.

Reciprocal loss-of-function confirmation: For candidate inhibitors demonstrating variant-dependent sensitivity, we will perform the reciprocal experiment in AA-derived PDO lines that endogenously carry the variant at the highest natural allele frequency. CRISPR-mediated knockout or reversion to wild-type will be performed using the same nucleofection and HDR approach, with reversion confirmed by sequencing.

Mechanistic characterization of validated candidates: For candidates demonstrating bidirectionally confirmed variant-dependent sensitivity, PDO cultures will be treated with vehicle or one of two concentrations (IC50 or IC75) before assessing cell viability, proliferation (EdU incorporation), apoptosis (Annexin-V staining), and expression of established CRC markers (CDX2, CK20, LGR5). Drug-specific biomarkers corresponding to the targeted ancestry-enriched driver will be validated by RT-qPCR or Western immunoblotting to confirm on-target activity and distinguish specific pharmacological effects from generalized cytotoxicity.

Future directions: While we do not formally propose organoid derivation from African-continent tumor tissue in the current application as we lack the preliminary feasibility data required to justify this as a formal aim, AfriCAN DANCE is designed to generate exactly the infrastructure needed to make this possible. Dr. Adjei and Dr. Adinku at KATH have demonstrated organoid derivation from fresh-frozen African tumor tissue in breast cancer^{65,66}, establishing proof of principle, and the prospective collection of fresh-frozen CRC specimens under AfriCAN DANCE will create the biospecimen resource from which African CRC organoids can be derived in a future application. Establishing African-continent-derived CRC PDO models is a high-priority next step that this study is specifically designed to enable.

Anticipated Outcomes: From the variant-stratified drug sensitivity screen, we anticipate identifying a subset of candidate inhibitors from Aim 2a with differential sensitivity between variant-carrier and non-carrier PDO lines, particularly among targets linked to the *MYC*, *WNT*, and *PI3K-AKT* signaling pathways enriched in our preliminary transcriptomic data. For high- and moderate-frequency variants including *KRAS G12D/G13*, *PIK3CA*, and *CTNNB1*, natural carrier counts among the 50 PDO lines provide 80–94% power to detect a two-fold IC50 difference, and we expect these tiers to yield the strongest primary screen signals. For low-frequency variants, CRISPR knock-in into NHW-background lines is expected to confer increased sensitivity to matched inhibitors relative to unedited isogenic controls, directly attributing differential sensitivity to the introduced variant. Reciprocal reversion experiments in AA-derived lines are expected to attenuate that sensitivity, providing bidirectional isogenic confirmation. The final output will be a set of variant-dependent druggable targets with on-target mechanistic validation, which can feed into future biomarker-stratified early-phase trials enrolling patients carrying the relevant variant regardless of self-reported ancestry.

Potential Pitfalls and Alternative Approaches: If the variant-stratified primary analysis yields weak differential sensitivity signals (e.g. top Aim 2a candidates reflect transcriptional rather than mutational drivers) we will shift the primary grouping variable from somatic variant carrier status to continuous African ancestry proportion, using the ADMIXTURE scores from Aim 1 to identify inhibitors whose activity tracks with ancestry proportion rather than any single variant. This approach sacrifices mechanistic specificity but preserves the ability to identify ancestry-associated therapeutic vulnerabilities from the same PDO dataset without additional experiments. If PDO lines do not endogenously express candidate drivers at sufficient levels to model functional impact, we will employ lentiviral overexpression to engineer expression of candidate genes in existing organoid lines, an approach previously validated in our laboratory⁶⁷ and expected to demonstrate oncogenic activity. This strategy preserves the ability to generate functionally validated targets regardless of endogenous expression characteristics and does not require generation of new primary cultures. If CRISPR knock-in efficiency is insufficient for a specific target locus, we will use base editing as an alternative precision editing strategy, which achieves single-nucleotide substitutions without double-strand breaks and has demonstrated higher efficiency than HDR-based knock-in at difficult loci in organoid systems.

Timeline: Shown in the Overall Research Strategy.